

AI-Ready: Preparing for and Using AI in Power BI and Microsoft Fabric

BELINDA ALLEN

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Thank You Sponsors!



3

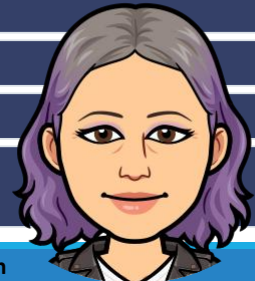


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 MsBelindaAllen

About Belinda Allen - Belinda@SACI.com

From Alcolu, SC	• I lived the longest in New York City
In Technology	• Enablement Advocate – Smith & Allen Consulting, Inc, <i>and</i> Principal Enablement Consultant– iLink Digital • Started in tech in 1991 • Microsoft Partner since 2001
Microsoft MVP	• 14 years / 5 years GP and 9 years Power BI
Microsoft MCT	• 15 years (with a hiatus in the middle)
Co-Authorred 2 books	• BI and Dynamics GP
I love learning	• Technology, Wood Turning, Cooking
I love hobbies	• Newest hobby: Pickleball • Favorite: Knitting/Crocheting



@MsBelindaAllen

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We are only reviewing Data Preparation for AI,
not AI itself (e.g. Data Agent, MCP Server, Etc.)

We only have 1 hour


6

<https://bit.ly/DDsatRIC26>


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Agenda

Copilot for Fabric and Power BI

Data Readiness

Copilot Prep

Stand alone Copilot

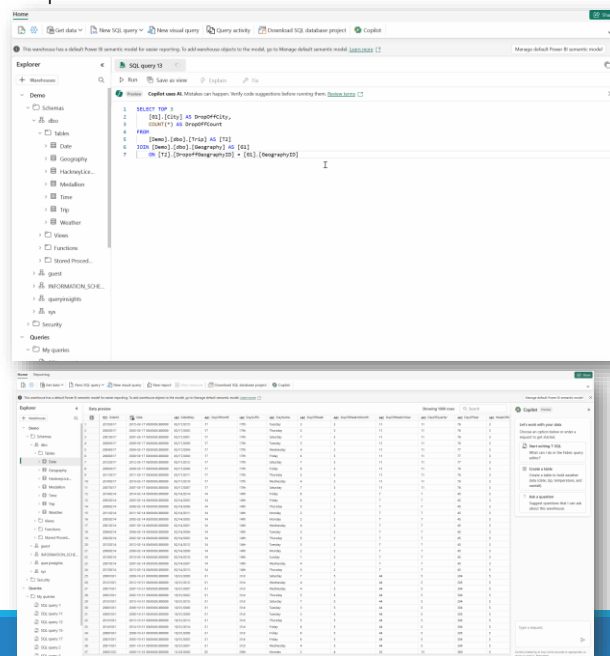
Limitations

Data Agents (Time permitting)

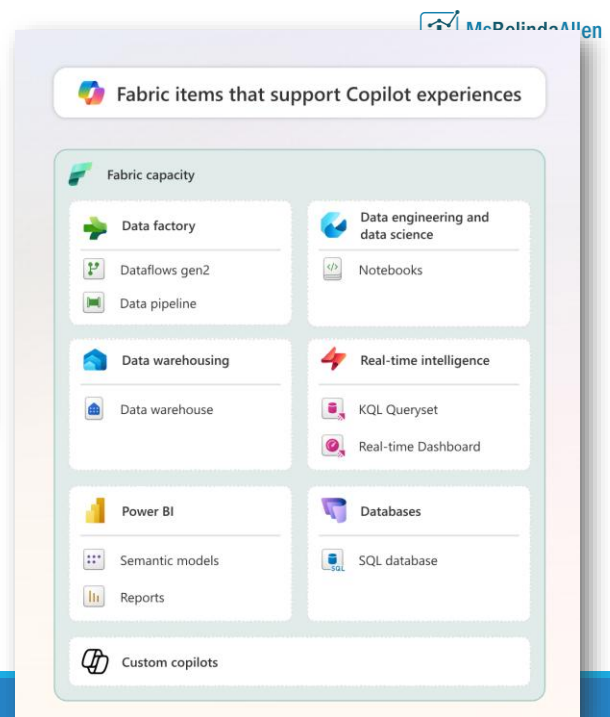
Q&A

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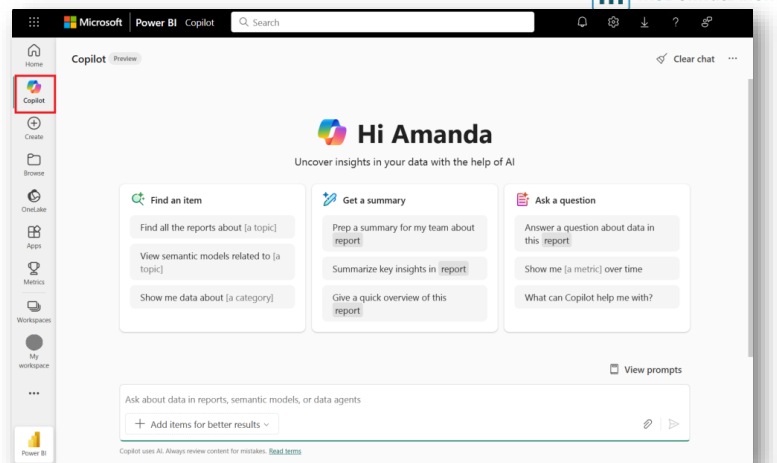
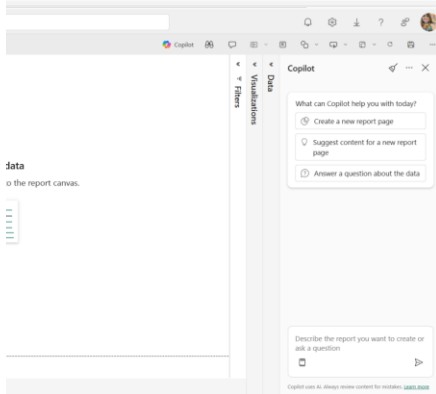
Copilot in Fabric



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Copilot in Power BI



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What is chat with your data? (CWYD)

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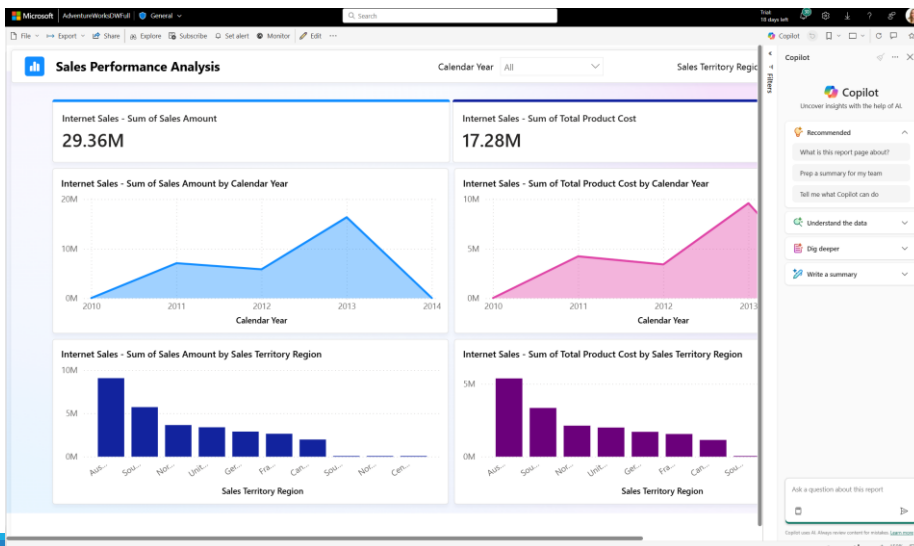
Where can I Chat with my data?

12



Where can you use Power BI Copilot?

The report pane



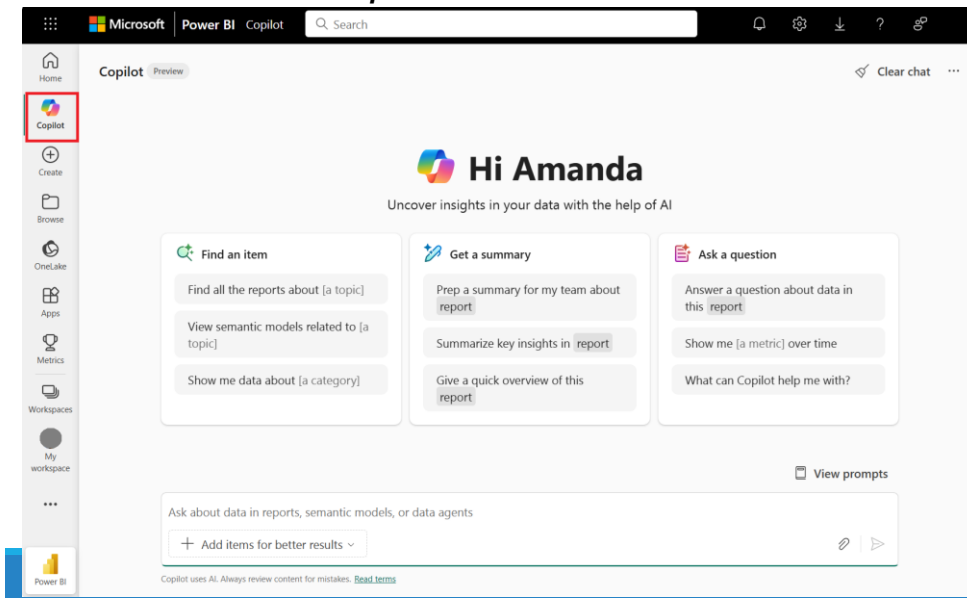
You can have Copilot give you...

- Summaries of the data in the report
- Answers from the report visuals
- Answers from the semantic model
- More detailed answers with ad hoc DAX generation

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Where can you use Power BI Copilot?

The standalone experience



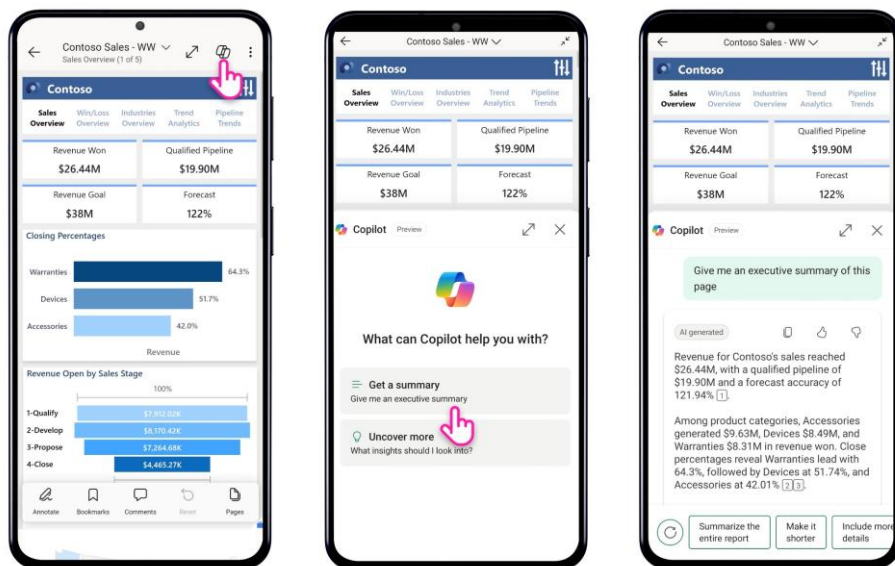
You can have Copilot ...

- Help you find relevant data
- Provide summaries of the data
- Answer questions from report visuals
- Provide insights from semantic models
- Generate ad hoc DAX queries
- Get answers from Data Agents

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Where can you use Power BI Copilot?

Power BI Mobile



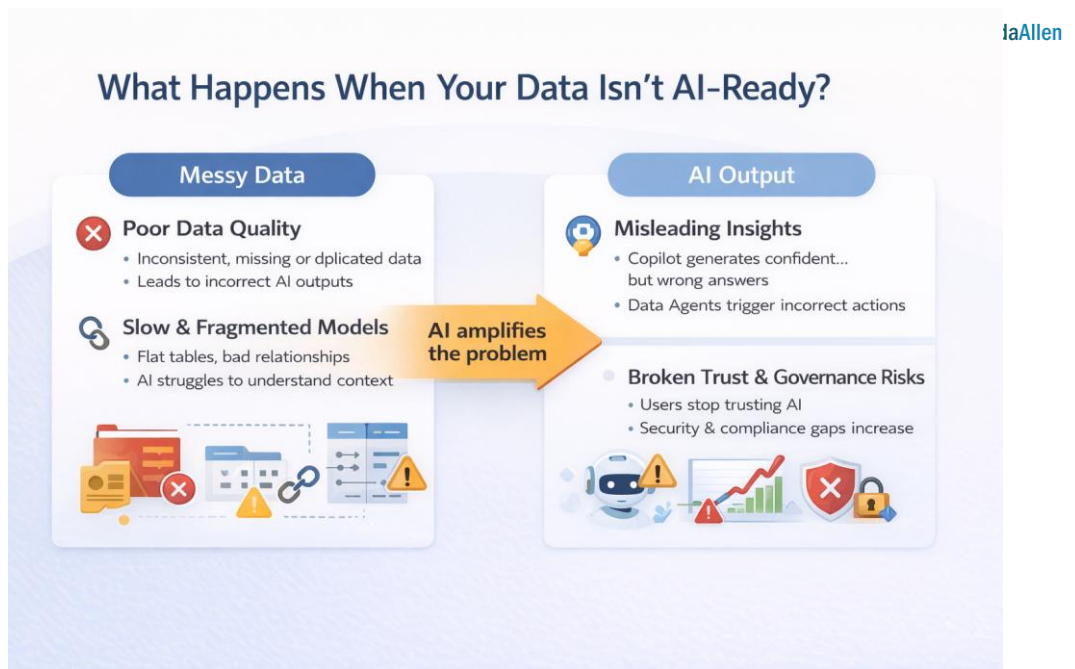
You can have Copilot ...

- Give a summary of data in reports
- Uncover additional insights from your data

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Data Readiness

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Word of the Day:

Non-Deterministic

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High-Impact Quick Wins

Business-Friendly Names

Rename tables, columns, and measures to plain business terms.

Avoid duplicates across tables.

Hide technical fields (IDs, keys, foreign keys, calc helpers).

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High-Impact Quick Wins

Model Structure

Favor a clean star schema.

Ensure relationships have correct cardinalities and intentional directions.

Remove inactive or ambiguous relationships where possible.


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High-Impact Quick Wins

Descriptions Everywhere

Add concise, business-first descriptions to tables, columns, and measures.

Include: what it is, when to use, units, and any baked-in filters.


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High-Impact Quick Wins

Explicit Measures

Create measures for every KPI and ratio; avoid implicit aggregates.

Use clear names (e.g., Sales, Sales LY, GM%, Active Customers).

Apply correct data types and formats (currency, percent, decimals).

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High-Impact Quick Wins

Date Intelligence

Use a single marked Date table.

Add time-based measures: YTD, QTD, MTD, WoW, YoY.

Include fiscal calendar fields if applicable.

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High-Impact Quick Wins

Data Categories & Summarization

Set categories: country, state, city, URL, email, latitude/longitude.

Set default summarization to None for IDs/text.

Choose appropriate defaults for numeric fields.

Use Sort by Column for ordered labels.

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High-Impact Quick Wins

Synonyms and Phrasing

Add synonyms/abbreviations for key entities and measures (e.g., revenue = sales, GM = gross margin, cust = customer).

Include regional variations (e.g., zipcode/postcode).

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High-Impact Quick Wins

Hierarchies & Display Folders

Create common drill paths:

- Date: Y–Q–M
- Product: Category → Subcategory → Product
- Geography: Country → State → City

Organize measures into clear folders.

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High-Impact Quick Wins

Performance Hygiene

Remove unused columns.

Reduce high-cardinality text where feasible.

Use incremental refresh for large fact tables for responsive answers.

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High-Impact Quick Wins

Governance & Trust

Apply row-level security where needed.

Set sensitivity labels.

Endorse the model to support trust in Copilot's answers.


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Demo


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High-Impact Quick Wins

Perspectives for Copilot

Create a “Copilot” perspective with only curated, business-friendly fields and measures.

Hide everything else to reduce ambiguity.

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Curated Copilot perspective example:

Dimensions

Date (Year, Quarter, Month)

Division

Age Group

Product / Item

Customer (if relevant)

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Curated Copilot perspective example:

Boutique – Top KPIs (Curated for Copilot)

Total Revenue by Year
 Total Revenue by Division
 Total Revenue by Age Group
 Total Purchases by Age Group
 Total Purchases by Division
 Total Revenue by Division & Age Group
 Purchase Count by Age Group
 Purchase Count by Division
 Best-Selling Items by Division and Year

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Curated Copilot perspective example:

Measures

Total Revenue
 Total Revenue by Year
 Total Revenue by Division
 Total Revenue by Age Group
 Total Revenue by Division and Age Group
 Total Purchases by Age Group
 Total Purchases by Division
 Purchase Count by Age Group
 Purchase Count by Division
 Best-Selling Items by Division and Year

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Curated Copilot perspective example:

Hide from Copilot

Keys, IDs

System columns

Technical fields

Columns not used in business analysis (e.g. measures for conditional formatting or titles)


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High-Impact Quick Wins

Verified Answers

Capture canonical answers for recurring questions (definitions, filters).

Add them as verified Q&A entries so Copilot prioritizes them.


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High-Impact Quick Wins

Test & Iterate

Ask real business questions.

Note where Copilot hesitates or misinterprets.

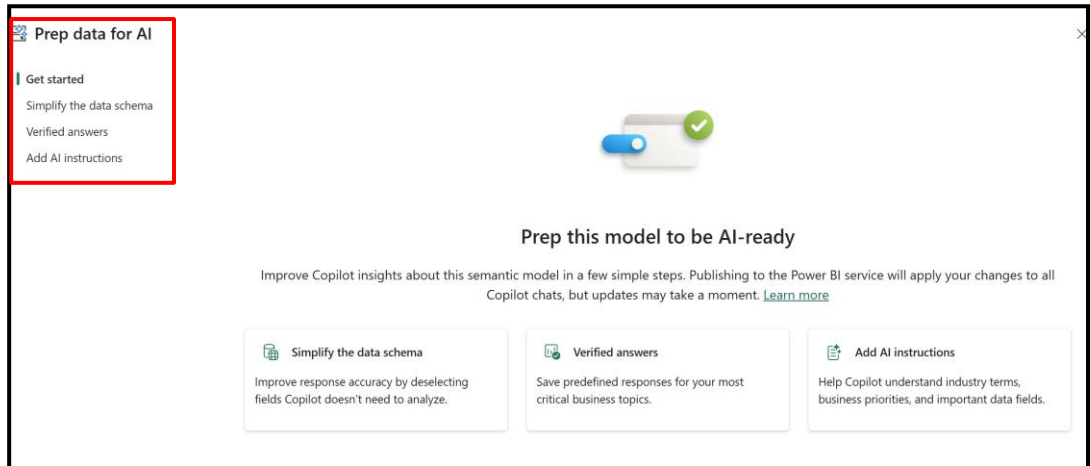
Add or adjust synonyms, descriptions, or measures accordingly.


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Copilot Prep


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Copilot Prep



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Copilot Prep – Instructions example:

If asked about "highest" or "most" or "best-selling" product, first clarify if the user wants product by unit sold or product by total sales value.

Best-selling = most units sold

Highest selling = total sales value

If you use an existing report visual to answer a user request, always let the user know about any existing filters on the visual.

If you do not understand what is being asked, do NOT guess, instead ask for clarification.

Visual Guidance

When showing the total sales measure by year always use a column chart.

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Q&A Tooling

Linguistic Schema:

- The metadata Power BI uses to interpret user queries.
- Includes synonyms, descriptions, and relationships.

Relationships:

- Define how tables connect (e.g., sales to products).
- Essential for Copilot to generate accurate DAX and visuals.

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Simplify the Data schema

Simplifying the Data Schema helps Copilot to focus on the right data by only exposing what Copilot needs to see

Visibility level	Original objects are still visible to users in reports
Scope of capability	Simplification only applies to Copilot and AI Features
Hidden state	All fields available in the semantic model appear in the AI data schema selection pane regardless of visibility set on the semantic model.
Hierarchies always respected	If a hierarchy is included in the AI data schema, but a column of the hierarchy isn't selected as part of the table, the column is still included as part of the Copilot schema.
Relationships always respected	Relationships are still respected regardless of the AI data schema set. For example, if two fields are related, and one of the fields is included in the AI data schema, Copilot can still return answers that require that relationship.
No end-user control	End-users can't disable the AI data schema.
Not a security feature	Do not use AI data schema as a security feature, it's a suggestion to Copilot for the schema to use when answering questions

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Verified Answers

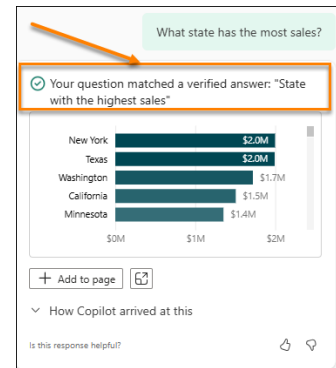
Verified Answers are approved, visual responses stored in the semantic model that Copilot can return when users ask predefined or semantically similar questions.

What do Verified Answers do?

- Provide trusted Responses
- Ensure Consistency
- Improve Copilots accuracy

How do they work?

- Select an existing visual and provide trigger phrases
- Copilot checks user prompts for exact or similar responses
- If matched, Copilot returns the verified answer, with filters and formatting



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Verified answers – Tips and tricks

Tip # 1 – Trigger phrases

Use exact and semantically similar trigger phrases – Provide 5 - 7 representative triggers per verified answer to cover different but valid ways users may ask the same question.

Tip # 2 – Semantic matches

Keep semantic matches realistic. Synonyms, word order changes, and filters in prompts work, but adding/removing fields or swapping measures does not.

Tip # 3 – Supported visuals

Select supported visuals and models – Verified answers only work with supported visual types (no AI visuals, decomposition tree, etc.) and supported models (Import, Direct Query, Composite, Direct Lake in service).

Tip # 4 - Visibility

Ensure fields and filters are visible – Hidden fields or locked filters break verified answers. Make sure schema selection includes all fields you expect Copilot to use.

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Verified answers – Tips and tricks (Continued)

Tip # 5 - Filter configuration

Configure filters correctly – Filters must be set to “All” in the report, visible to end users, and within the supported types (basic categorical, some advanced categorical, date range). Limit to 10 filters during creation, 3 at runtime



Tip # 6 - Understand answer persistence

Verified answers are tied to the semantic model, not the original report visual. Editing or deleting the report visual doesn't update the verified answer—manage updates in the Verified Answer setup dialog.



Tip # 7 - Test carefully with the skill picker

In desktop, disable authoring skills like “create page” to mimic end-user Copilot behavior and validate your verified answers trigger correctly.



Tip # 8 - Communicate clearly with users

Verified answers show a checkmark, matched phrase, and “how Copilot arrived at this.” Make sure these elements are accurate and trustworthy for confidence building.

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Verified answers – Tips and tricks (Continued)

Tip # 9 - Stay within limits

Each model supports up to 250 verified answers, 15 trigger phrases per answer, and 500 characters per trigger. Respect these caps for reliability.



Tip # 10 - Publish and validate in service

After creating verified answers in desktop, publish to service and allow ~15 minutes for them to appear. A small edit in service (like tweaking a trigger phrase) can speed up sync



11. Select the visual first

The “Set up Verified Answer” option only appears when a visual is selected.



12. Manage through the Verified Answers dialog

Edits to report visuals won't update Verified Answers; use the management dialog or recreate them.

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AI Instructions

AI Instructions give business context and instructions for Copilot to use to provide better answers

What do AI Instructions do?

- Provide context and guidance to Copilot
- Define terminology, tone, and response style
- Help Clarify business terms and guide analysis approaches

Why AI instructions?

- AI Instructions align Copilot with your business
- They improve the quality and relevance of responses
- More meaningful insights, less user frustration, smoother experience

Testing and Validating

Testing AI instructions will be faster in Power BI Desktop because there is no publishing delay. We currently* recommend testing and refining your AI Instructions in Power BI Desktop

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AI Instructions Best Practices

Be explicit and specific

- Assume Copilot has no understanding of how to use the data model or business context for your data

Be Descriptive

- Use Analogies and descriptive language to help the model understand your desired outcome.

Avoid ambiguity

- If something should be emphasized - or avoided - be clear. The more details and supporting content you can provide Copilot, the better.

Group related instructions

- Organizing instructions by theme or purpose can help Copilot better understand your intent.

Order Matters!

- AI Instruction order affects Copilot's response, test different sequences and phrasing for best results.

Break down complex instructions

- Break down complex instructions into simpler steps to improve clarity and reduce errors

Keep instructions focused

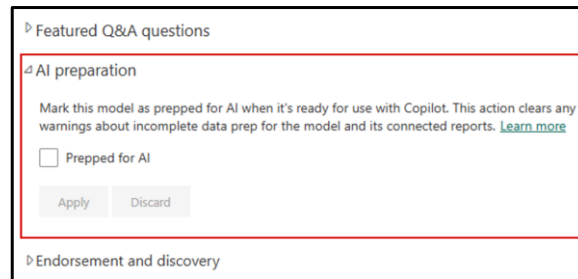
- Too many or conflicting instructions can confuse the model, sometimes less is more.

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Prepped for AI Setting

Marking a semantic model as **Prepped for AI** tells Copilot it's a good item to use.

We boost prepped items in the search list, and we won't show content warnings on answers that use prepped data.

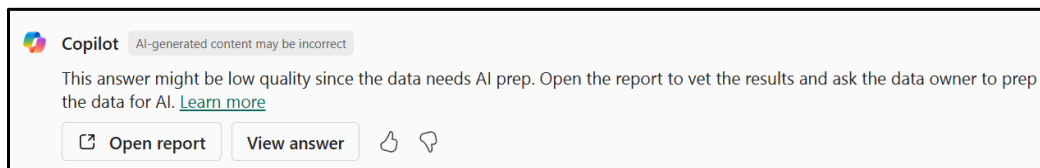


Find this switch in the **AI preparation** section of semantic model settings

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Warnings on unprepped data

Copilot provides content warning on answers that use data from an unprepped semantic model



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Demo 2

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Copilot in the Standalone experience

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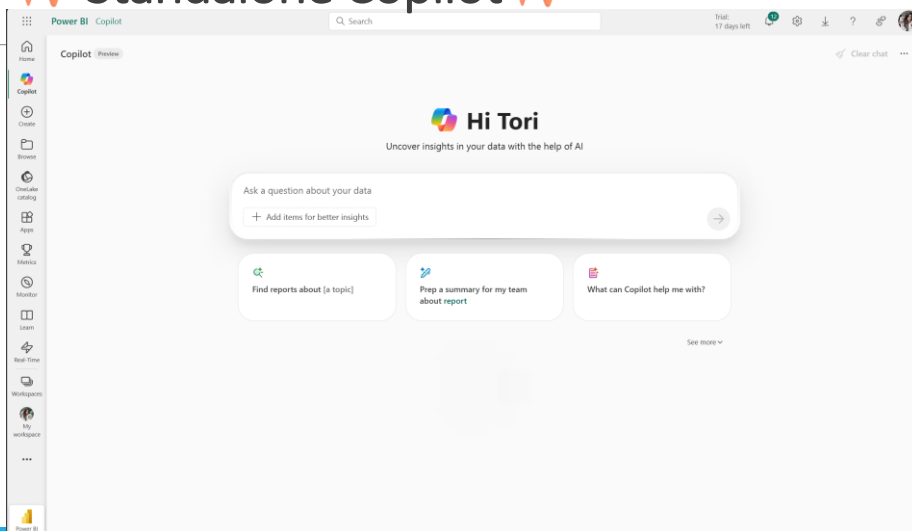
Lost in workspaces? Copilot finds a way

Organizations today face report sprawl, thousands of visuals across countless workspaces making data insights challenging and time consuming

How can users look through all their artifacts and find answers to their data questions quickly?

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Enter ✨ Standalone Copilot ✨

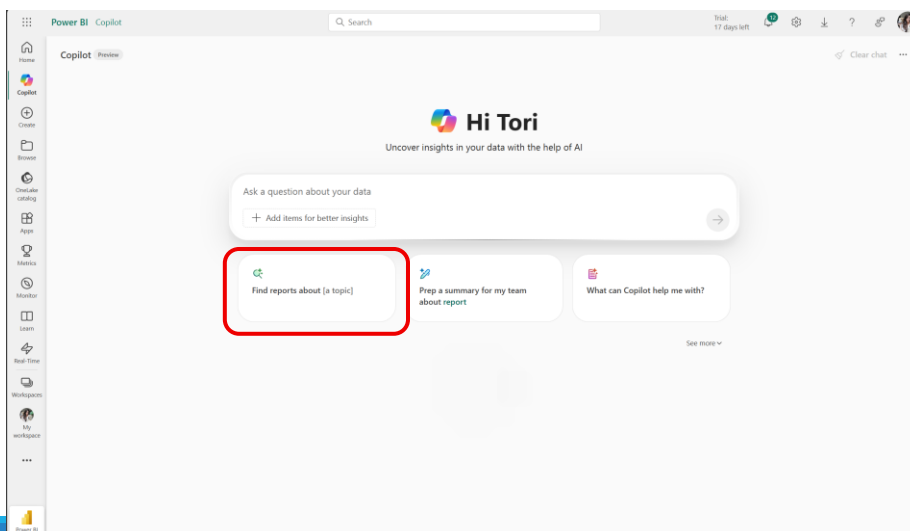


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Standalone Copilot capabilities

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Capability: Search



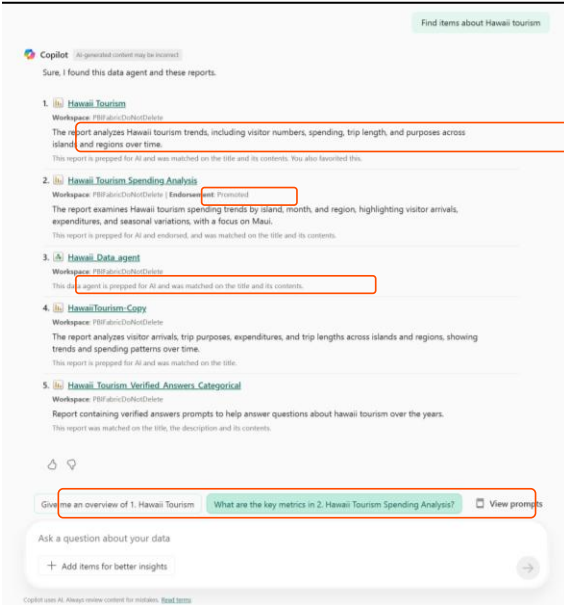
Find the right:

- ✓ Models
- ✓ Reports
- ✓ Data Agents
- ✓ Apps

Trigger with a Search question or data question without a report attached

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Copilot can help you find data content on a topic



Items are linked to open in new tabs

LLM-gen captions describe reports

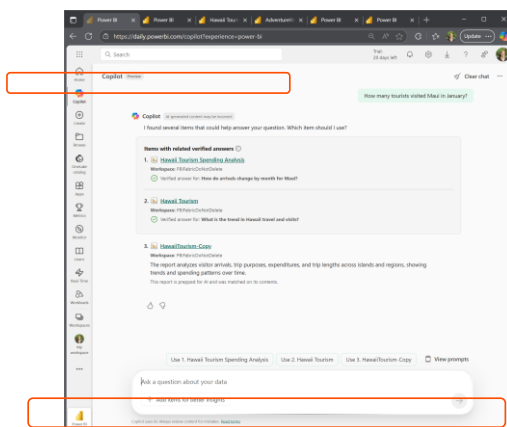
Core item metadata (like workspace & endorsements)

Context on why this item was returned

Suggested questions for next actions with Copilot

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Or ask a question, and it will suggest items to answer



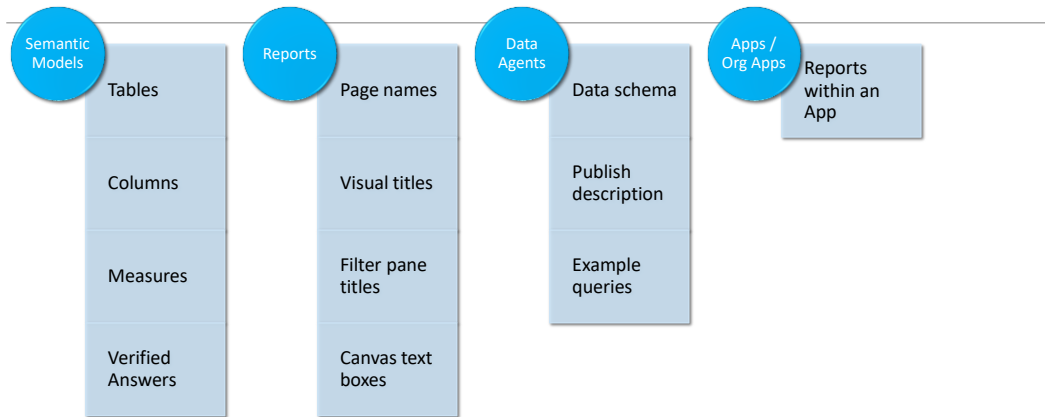
Related **verified answers** are surfaced at the top of the list

To select an item to answer from, the pills at the bottom can be used, or the user can type "Use 1"

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What can Copilot search?

All content that the user has access to in Power BI:



Item-level metadata: Title, Description, Endorsements

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How is search content boosted?

Curation signals

- Prepped for AI
- Verified answers
- Promoted
- Certified

Usage signals

- Personal views (last month)
- Personal Favorites
- Popular in your org

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How do I make items more discoverable?



Add item
descriptions



Use unique
keywords



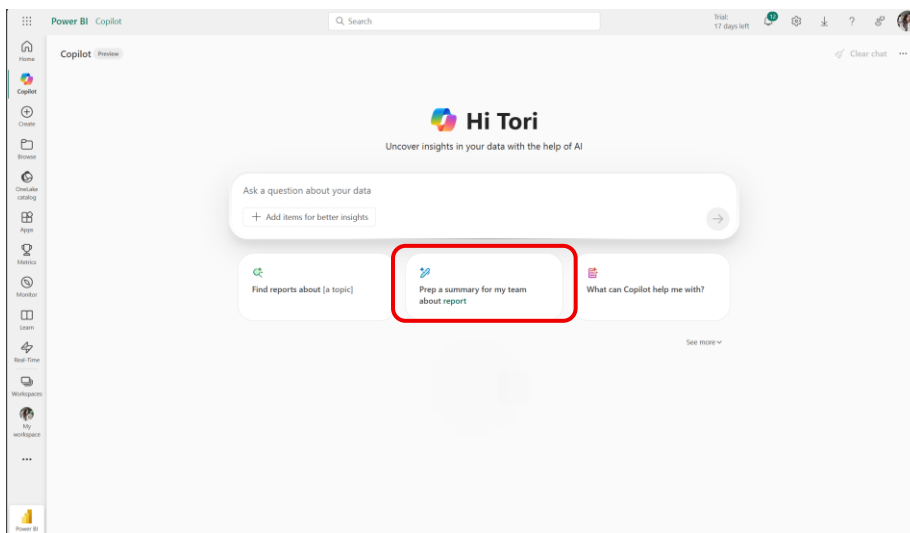
Create **Verified**
Answers for
FAQs



Prep or
endorse

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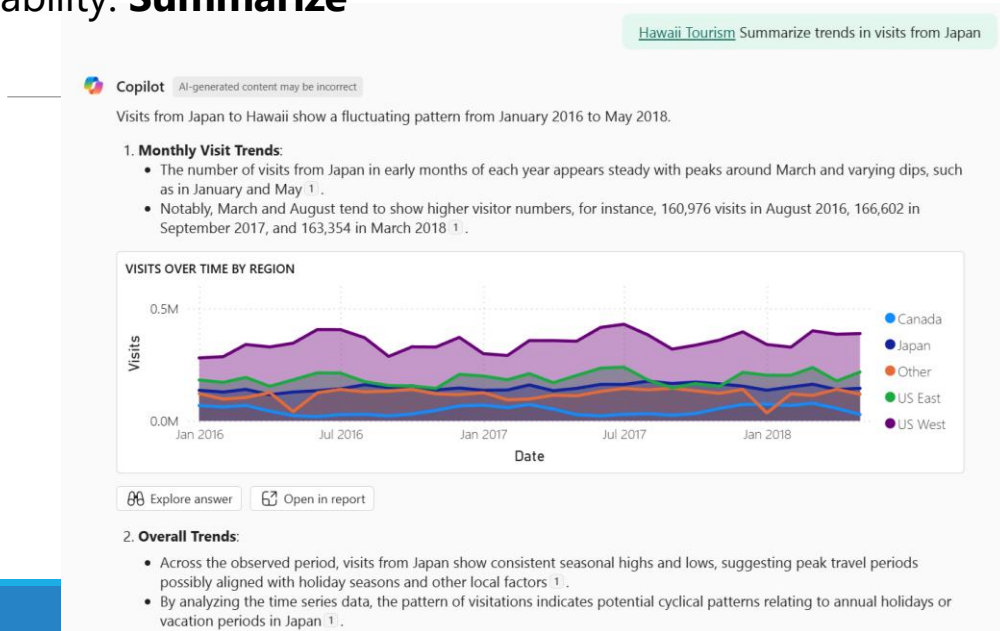
Capability: **Summarize**



Summarizes
reports or
specific topics
in reports

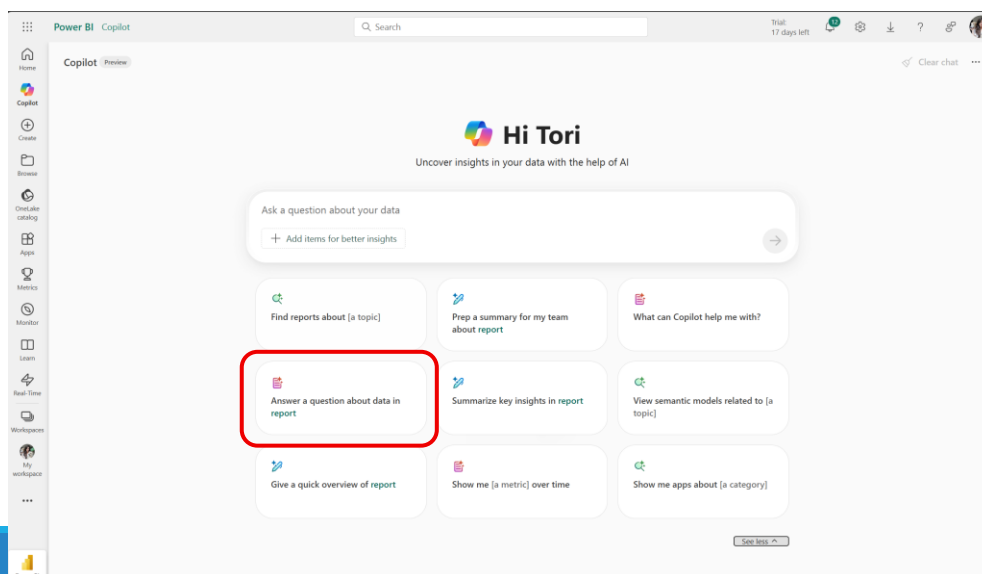
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Capability: Summarize



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Capability: Answer Data Questions

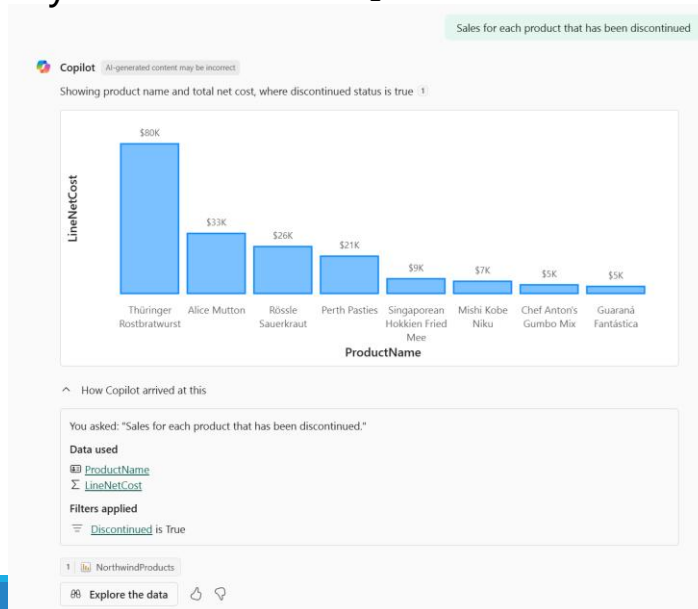


Can find answers within existing report visuals

Can generate new analysis from semantic model data

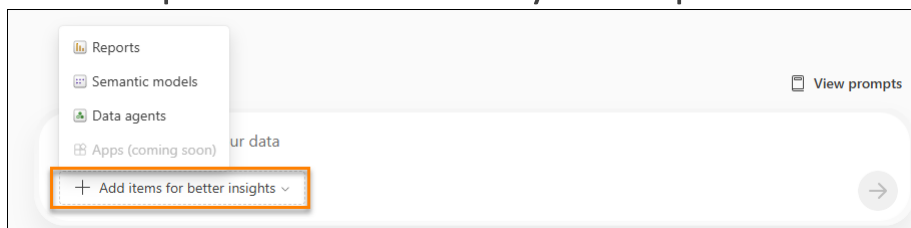
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Capability: **Answer Data Questions**



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Attach specific data to your question



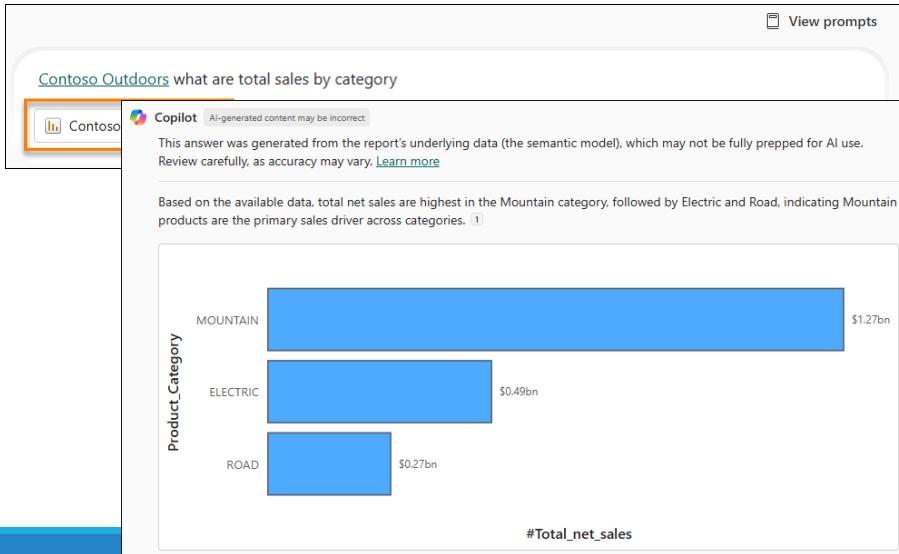
Ask questions on specific artifacts

Reports
Semantic Models
Data Agents
Apps (coming soon)

Use Fabric Data Agents without leaving standalone

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Attaching a report

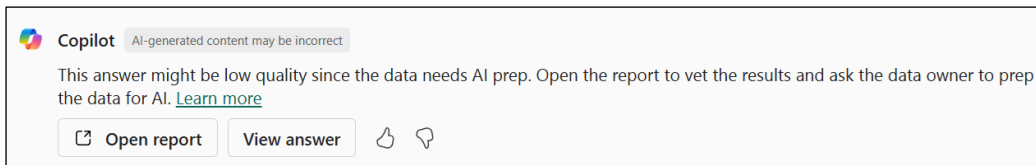


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Friction Treatment

Unprepped data can lead to low quality answers

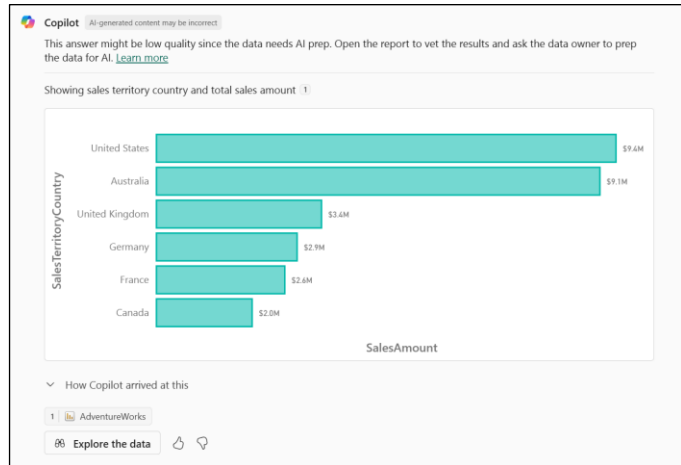
Standalone Copilot provides a disclaimer on answers to questions over an unprepped semantic model



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Friction Treatment

Users must select the View Answer button to reveal the full answer



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Demo 3

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Technical & Functional Limitations

Preview Feature:

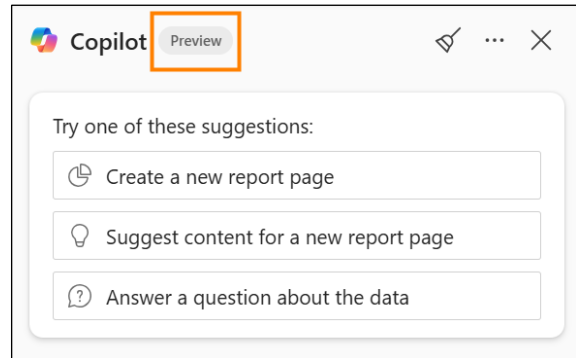
The standalone Copilot experience is still in preview and not available in all regions

Data Prep Required:

Semantic models must be marked as "AI-prepped" to avoid low-quality answers

Limited Support in Embedded Scenarios:

Copilot is not supported in "App Owns Data" embedding models



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Human Oversight Still Needed

Not a Replacement for Expertise:

Copilot can assist but not replace human intelligence for complex modeling or nuanced analysis

 **Mistakes are possible**
Carefully review content created by AI before using it.

Requires Foundational Knowledge:

Users still need basic understanding of DAX, data modeling, and Power BI concepts



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Scope of Capabilities

Report-Only Summarization:

Summaries are currently limited to reports, not semantic models

Accuracy Caveats:

Generated insights and DAX formulas should be reviewed for correctness

create a dax measure for rank of cities by total banks

I can't create DAX measures directly, but I can guide you on how to write a DAX measure for ranking cities by total banks. Here's a sample DAX formula you can use in Power BI:

```
City Rank by Total Banks =
RANKX(
ALL('Failed Banks'[City ]),
CALCULATE(COUNTROWS('Failed Banks')),
,
DESC,
DENSE
)
```

This measure will rank each city based on the total number of banks, with 1 being the city with the most banks. You can add this measure to your Power BI model by creating a new measure in the 'Failed Banks' table. If you need further help visualizing or using this measure in your report, let me know!

Is this response helpful?




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Fabric Data Agents (Time Permitting)

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Why Data Agents?

Curated Insights

Focus conversations on specific datasets—no noise from unrelated models.

AI-Powered Querying

Natural language interface—no need for SQL, DAX, or KQL.

Multi-Source Connectivity

Combine semantic models, lakehouses, and warehouses in one agent.

Customizable Behavior

Use AI instructions to guide tone, format, and data source priority.

Shareable & Discoverable

Publish agents across Fabric, Teams, Copilot Studio, and custom apps.

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Data Agents vs. Standalone Copilot

Feature	Data Agents	Standalone Copilot
Scope	Curated to specific datasets	Broad access across all workspaces
Customization	Highly customizable with AI instructions	Limited customization
Data Source Control	Select specific tables and sources	Uses all available semantic models
AI Behavior Tuning	Supports tone, format, and source priority	No AI instruction support
Use Cases	Targeted insights, specialized scenarios	General exploration and discovery
Integration	Shareable across apps (Teams, Copilot Studio)	Embedded in Fabric workspace

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Add AI Instructions

Why Add AI Instructions?

- Use AI Instructions to better align the Fabric data agent with your organizations needs, ensuring more accurate and context aware responses.

Where?

- AI instructions can be added at the agent level under Agent instructions or at the data source level under data source instructions

Limitations

- You can write up to 15,000 characters in plain English-language text.
- Currently semantic models are not supported.

Adding AI Instructions

AI instructions

Agent instructions

Guide the data agent to the best data sources for different types of questions by writing a detailed system prompt.

Set Response guidelines

Always include a concise summary before the detailed breakdown.
Always return a table instead of bullet points

Summary: Sales figures are only available for one country in the data.

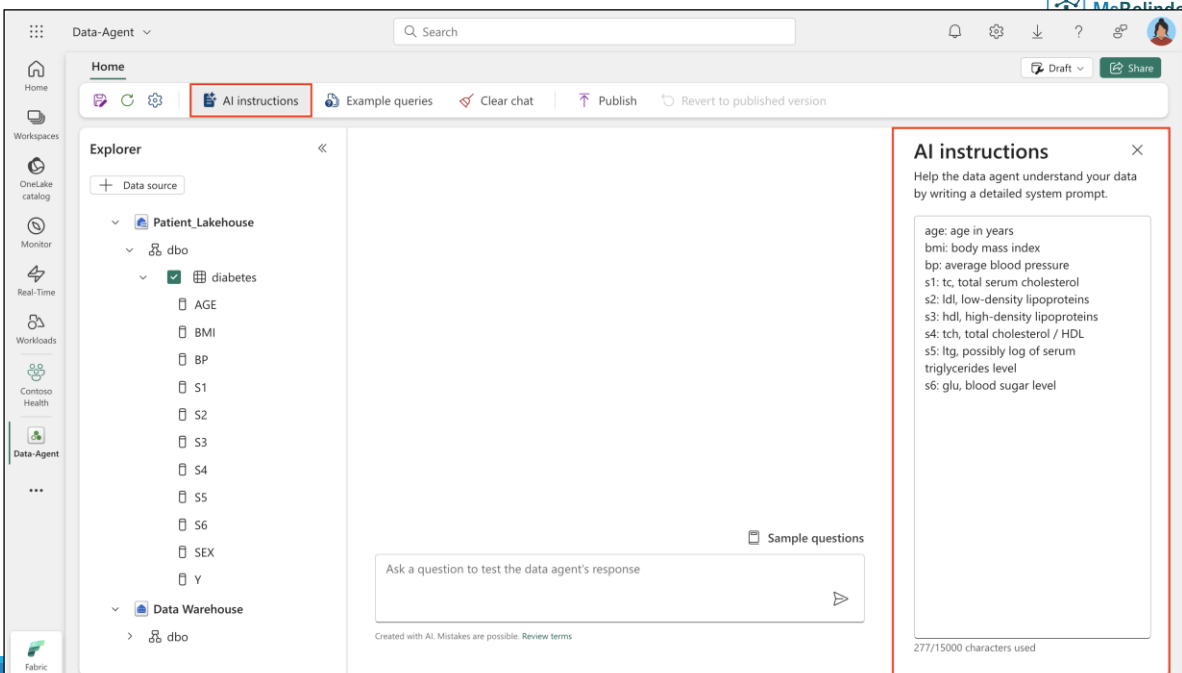
Table:

Country Name	Total Sales
United States	\$105,724,059.25

1 step completed

Response time: 18 sec

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The screenshot shows the Microsoft Fabric Data-Agent interface. On the left, the 'Explorer' pane shows a hierarchy of data sources: 'Patient_Lakehouse' (containing 'dbo' and 'diabetes') and 'Data Warehouse' (containing 'dbo'). The 'AI instructions' tab is selected in the top navigation bar. The main area displays a text input field for AI instructions, with a red box highlighting the instructions: 'age: age in years', 'bmi: body mass index', 'bp: average blood pressure', 's1: tc, total serum cholesterol', 's2: ldl, low-density lipoproteins', 's3: hdl, high-density lipoproteins', 's4: tc, total cholesterol / HDL', 's5: ltc, possibly log of serum triglycerides level', 's6: glu, blood sugar level'. Below the input field, a 'Sample questions' section is visible. The bottom right corner shows '277/15000 characters used'.

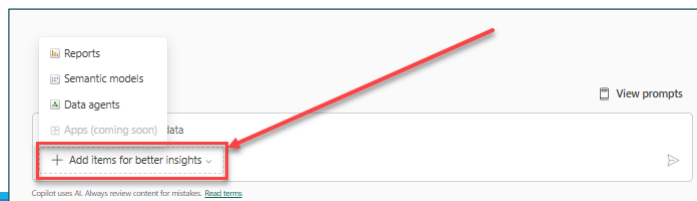
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Consuming Data Agents

The published Fabric data agent can be consumed in various platforms, including:

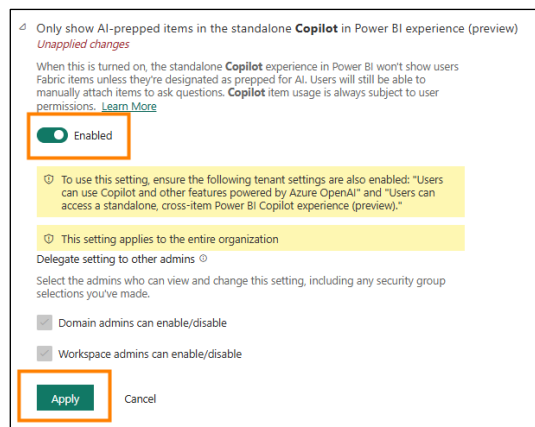
- Microsoft Fabric**
- Copilot Studio**
- Microsoft Teams**
- Power BI Copilot**
- Azure AI Foundry**
- Custom applications via API**

These integrations enable conversational AI, allowing users to query data in natural language and retrieve insights.



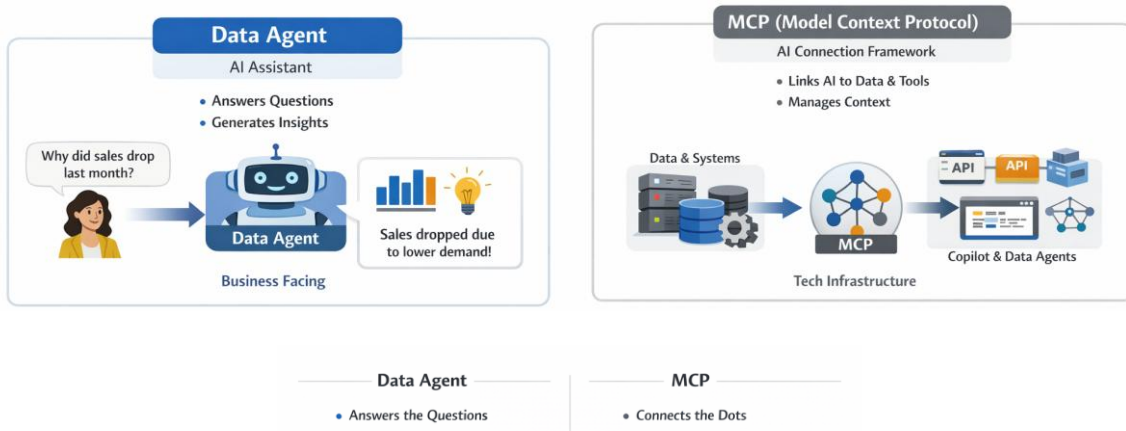
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Only show AI-prepped items in the standalone Copilot in Power BI experience



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Microsoft Fabric: MCP vs. Data Agent



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Demo 4 (Time Permitting)

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Day of Data
Richmond

Thank You Sponsors!



82



POWERBI DAYS DC

June 11-12 2026
www.powerbidc.org



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Thank you!

Q&A



<https://bit.ly/DDsatRIC26>